

The Sweet Lie Theorem

The Möbius quantization of the Fermat fourfold rank — the families lip-sync; the song belongs to the stage

Rafael Amichis Luengo · Madrid · tretoef@gmail.com · github.com/tretoef-estrella · 6 June 2026 · Companion to THE_WATERMARK_THEOREM and THE_DOUBLE_LADDER_THEOREM (same repository). Computations: MacBook Air M2, 8 GB RAM, single thread, 25% CPU.

Abstract

For the Fermat fourfold $X_d \subset \mathbb{P}^5$ of degree $d \geq 3$, let V be the lattice spanned by the classes of its $15 \cdot d^3$ standard planes — fifteen families, one per perfect matching of the six coordinates, with d^3 translates each. For a subset S of the families define the Möbius interaction mass $v(S) = \sum_{T \subseteq S} (-1)^{|S \setminus T|} \text{rank } V(T)$. We prove:

Theorem (Sweet Lie). Let G_S be the union graph on the six coordinates of the matchings in S , with components classified bipartite / non-bipartite, and let $N_d(G_S)$ be the product over components of $(d-1)$ for a bipartite component and ε_d for a non-bipartite one, where $\varepsilon_d = 1$ if d is even and 0 if d is odd. Then for every nonempty S and every degree $d \geq 3$, prime or composite:

$$v(S) = (-1)^{|S|+1} \cdot (1 + N_d(G_S)).$$

Telescoping over all 2^{15} subsets yields, in both parity branches as polynomial identities,

$$\text{rank } V(4,d) = 15d^3 - 90d^2 + 175d - 99 + \delta \cdot (15d - 39), \delta = (d-1) \bmod 2,$$

which equals $DS_4(d) + 1$, the rank value of Degtyarev–Shimada's Remark 4.4 plus the hyperplane class. The theorem hands every constant of that formula to a finger count over the geometry of K_6 — fifteen matchings, forty-five pencil and sixty skew pairs, an alternating binomial sea, and ten bipartite hexads (the matching-sets of the ten coordinate bipartitions, each a $K_{3,3}$) — and identifies the mechanism: **resonance is bipartiteness**. The name records the discovery that drove the proof: the families appear to sing, but every resonant dimension they exhibit belongs to a coordinate bipartition — the stage — and in even degree one global impostor character sings in every choir at once. Every law below was measured before it was proven, five complete kingdoms (all 156 subset-types at $d = 3, 4, 5, 6, 7$ — 775 nonempty type-measurements) anchor the statement with not one off-formula value, the even-degree corollary was declared from the proof and then confirmed blind, and the graveyard of dead candidate laws is published with its falsifiers as part of the record.

1. The question and the answer

Degtyarev–Shimada [1] proved, via their criterion, a closed rank formula for the lattice generated by the planes of the Fermat fourfold (their Remark 4.4); Aljovin–Movasati–Villafior [4] computed the lattices cell by cell within their hardware's reach (their fourfold rows stop at $d = 6$, at costs reaching 170 GB of swap for a single cell). What no source provides is the **interaction structure** behind the formula: how the rank of an arbitrary alliance of plane families is assembled, which alliances carry more than their parts, and why the formula's coefficients are the integers they are. The present theorem answers this by computing the full Möbius spectrum of the rank function on the lattice of family subsets — a 2^{15} -point object compressed, by S_6 -symmetry, to 156 types — and finding it **quantized**: for odd degree, every alliance of three or more families interacts through either exactly one shared dimension or exactly d , nothing in between; the alliances that carry d are precisely those whose combinatorial footprint is bipartite. The rank formula of [1] then falls out by telescoping, with every coefficient a count over K_6 .

Three layers of evidence anchor the theorem beyond the proof. **Pointwise**: complete ladders — every one of the 156 subset-types — at five kingdoms, $d = 3, 4, 5, 6, 7$, all 775 nonempty values on-formula, with the $d = 4$ and $d = 6$ ladders (the even branch and the first composite degree) measured **after** the formula was written. **Blind**: the $d = 11$ spot trial (4/4 sealed to the digit) and the even-degree corollary's first prediction (§5), a value no odd-degree measurement could have suggested, declared from the proof and then measured. **Symbolic and independent**: both telescope branches verified as polynomial identities, plus a from-scratch independent rebuild of the formula (Auditor environment) reproducing $DS_4(d)+1$ numerically at all thirteen degrees $3 \leq d \leq 15$ — including degrees whose lattices have never been computed by anyone.

2. Notation and substrate

$H^4_{\text{prim}}(X_d) \otimes \mathbb{C}$ decomposes as $\oplus_a \mathbb{C} \cdot v_a$ over characters $a = (a_0, \dots, a_5)$, $a_i \in \mathbb{Z}/d \setminus \{0\}$, $\sum a_i \equiv 0 \pmod{d}$; the coordinate torus μ_{d^6}/diag acts on v_a by a [2]. The fifteen families of standard planes correspond to the perfect matchings of the six coordinates; the d^3 planes of a family form a single orbit of its edge-torus. Two facts of the classical Fermat machinery are used as substrate: the eigenbasis description above and its torus equivariance (textbook), and a **Nonvanishing Lemma** — *the class of a standard plane has nonzero coefficient on every character of its support* — a Shioda-type computation (plane classes have product-of-Jacobi-factor coefficients in the eigenbasis [2, 3]). The Lemma's status is stated plainly in §6.

3. The proof — the support calculus

(a) **Master formula.** Averaging a plane class over a subtorus kills every character nontrivial on it; with the Nonvanishing Lemma, the span of one family F (matching M) is $\langle h \rangle \oplus \text{span}\{v_a : a_x \equiv -a_y \text{ for every edge } (x,y) \in M\}$ — the **character support** of F , plus the hyperplane class. Spans of alliances are sums of spans, so

$$\text{rank } V(S) = 1 + |\cup_{F \in S} \text{supp}(F)|.$$

(b) **Möbius duality.** The Möbius transform of $|\cup|$ over the subset lattice is $(-1)^{|S|+1}|\cap|$, and of the constant 1 it is $(-1)^{|S|+1}$. Hence

$$v(S) = (-1)^{|S|+1} \cdot (1 + |\cap_{F \in S} \text{supp}(F)|).$$

(c) **Counting the common support.** A character a lies in every support of S iff $a_x \equiv -a_y$ along every edge of the union graph G_S — iff a is an alternating $\pm r$ labeling on each connected component. A bipartite component admits $d-1$ such labelings ($r \in \mathbb{Z}/d \setminus \{0\}$); a non-bipartite component contains an odd closed walk, forcing $2r \equiv 0$, whose only nonzero solution $r = d/2$ exists iff d is even. Multiplying over components gives $|\cap \text{supp}| = N_d(G_S)$. The condition $\sum a_i \equiv 0$ is automatic (matched values cancel in pairs), and every such character is Hodge: each value pair $(x, -x)$ contributes fractional-part sum 1 at every unit t , and three pairs give exactly 3. ■

(d) **The corollaries this absorbs.** $K = 1$: a matching is three disjoint edges, three bipartite components, $N = (d-1)^3$ — the flat law $\text{rank } (d-1)^3 + 1$, which this campaign verified at ten degrees (eleven family-tower verifications — $d = 10$ at both primes). $K = 2$: a skew pair's union is a single 6-cycle (bipartite), $N = d-1$, so $v = -d$; a pencil pair is a doubled edge plus a 4-cycle, $N = (d-1)^2$, so $v = -((d-1)^2 + 1)$ — both verified at four kingdoms before the proof existed. $K \geq 3$, odd d : $N \in \{0, d-1\}$ — the **quantization**, ± 1 or $\pm d$ and nothing between; $N = d-1$ exactly when G_S is connected bipartite, which happens precisely when S lies inside the matching-set of one coordinate bipartition. Those matching-sets are the **bipartite hexads**: ten of them (one per 3+3 split), each the six perfect matchings of a $K_{3,3} \subset K_6$, each isomorphic to S_3 , no subset of size ≥ 3 lying in two — so the resonant masses at $K = 3, 4, 5, 6$ are the finger counts $10 \cdot C(6, K) = 200, 150, 60, 10$, and nothing of size ≥ 7 resonates.

(e) **The choruses and the universal note (the mechanism, exhibited).** For a bipartition β , the $d-1$ alternating characters span, with $\langle h \rangle$, a d -dimensional space D_β — the **chorus** — contained in the span of each of the hexad's six families and (for odd d) of no other: a family outside has an edge inside one part, forcing $2r \equiv 0$. Interior alliances re-meet the same D_β , and inclusion–exclusion rebounds by $+d$: resonance is six families repeatedly meeting in one shared chorus. Two different choruses share only $\langle h \rangle$ — the **universal note**, present in all fifteen spans; its explicit representative is the famsum identity (the sum of a family's d^3 plane classes is one and the same vector regardless of the family — verified independently at two kingdoms, all fifteen families). Oriented charge sums of any single hexad family generate its chorus (closed form verified at two kingdoms); the song belongs to the stage.

4. The telescope — the formula as pure counting

Summing v over all nonempty subsets (Möbius inversion):

$$\text{rank } V(4, d) = 15[(d-1)^3 + 1] - 45[(d-1)^2 + 1] - 60d + 91 + 100(d-1) \text{ (d odd)}$$

with 15 families, 45 pencil and 60 skew pairs, $91 = \sum_{K \geq 3} (-1)^{K+1} C(15, K)$ the alternating generic sea, and $100 = 200 - 150 + 60 - 10$ the alternating resonant mass. Expansion gives $15d^3 - 90d^2 + 175d - 99 = DS_4(d) + 1$ identically. For even d the impostor's contributions (§5) add exactly $15d - 39$. Both branches are symbolic

identities, and the numerical telescope reproduces every measured cell of this campaign — 21, 142, 401, 1002, 1861, 3302, 5121, 7762, 10901 at $d = 3..11$ — and predicted rank $V(4,13) = 19921$ before that cell's first computation.

5. The impostor and his audited debt (even degree)

For d even the character $(d/2) \cdot (1,1,1,1,1,1)$ is Hodge and alternates along **every** edge of K_6 : one global impostor singing in every choir of every bipartition and in every family span. First consequence, declared from the proof and then measured blind at $d = 4$: the generic triple whispers $v = +2$ (not $+1$) — together with $v(\text{skew}) = -4$, $v(\text{pencil}) = -10$, $v(K_{3,3}\text{-triple}) = +4$, four predictions, four hits, independently reproduced. Second consequence, the even-degree surcharge of DS audited céntimo a céntimo: summing $(-1)^{|S|+1}(d-1)^{|b(S)|}$ over the alliances with at least one non-bipartite component, the only d -scaling line is the fifteen pencil triples (a bipartite doubled edge paying $d-1$ beside a K_4 casino where only the impostor sings), contributing $15(d-1)$; the pure casinos' alternating ledger sums to -24 ; total **$15d - 39$** — exactly DS's δ -term, identically.

6. What is proven, what is verified — the honest ledger

The support calculus of §3 is complete modulo two classical inputs: the eigenbasis description of Fermat middle cohomology with its torus equivariance (textbook, [2]), and the Nonvanishing Lemma of §2 — a Shioda-type Jacobi-factor computation, **cited-route**, and empirically enforced by every measured dimension of this campaign: any vanishing anywhere would have dented a dimension somewhere across five complete kingdoms, the ten-degree flat law, and the blind trials. Pinning the precise literature citation (or writing the two-line Jacobi-factor verification in full) is the one open item this document holds itself to before submission, and is stated here as exactly what it is. The discriminant-group profiles of the cells (the torsion floors) are **outside** this theorem — they are the campaign's next object, and nothing here depends on them.

7. Provenance and discipline of record

DS [1] proved the rank formula by their criterion; AMV [4] computed cells within their tables' reach. New here: the Möbius quantization, the bipartite hexads and their choruses, the character-support mechanism, the even-degree impostor with its audited debt, and the derivation of the formula as a finger count over K_6 . The operation ("Légolas", commissioned in the campaign's findings master §86–§93) ran under a standing dual-instance protocol (Constructor / Auditor, the Architect as courier): every instrument reproduced in a second independent environment before ratification; every law declared before measurement wherever a new cell was touched; sealed blind predictions at every step (the (4,9) and (4,8) rank seals, the $d = 11$ triple seal, the $d = 4$ even colleja, the (4,13) rank) — and the (4,7) graveyard that set the zero-profile-commitment discipline every later seal obeyed.

The graveyard is part of the record, published with pride — each grave bought a road: the prime ladder $h_{\max} = (d+1)/2$ and $E = 3(d-2)^4 + 6d(d-3)$ (3-point laws, executed by the fourth kingdom $d = 11$ with falsifiers $4 \neq 6$ and $15603 \neq 20211$); cubic per-floor forms (killed by pencil: half-integer interpolants at $d = 13$); per-character floor statistics (killed by a proven no-go: one transitive orbit versus three measured levels at $d = 3$); subset-cumulative floor reconstruction (alternating divergence at two kingdoms); "born at $K = 5$ " (killed by the complete type sweep); $v(\text{pencil}) = 13 - 6d$ (a 2-point fit, dead at its third kingdom); two premature refinements of the charge-sum fine structure (dead at $d = 5$). Each falsifier is in the findings master with the data that fired it.

References

1. A. Degtyarev, I. Shimada, *On the topology of projective subspaces in complex Fermat varieties*. J. Math. Soc. Japan 68:3 (2016), 975–996. arXiv:1405.4683.
2. T. Shioda, *The Hodge conjecture for Fermat varieties*. Math. Ann. 245 (1979), 175–184.
3. N. Aoki, T. Shioda, *Generators of the Néron–Severi group of a Fermat surface*. Progress in Mathematics 35, Birkhäuser (1983), 1–12.
4. E. Aljovin, H. Movasati, R. Villafior, *Integral Hodge conjecture for Fermat varieties*. J. Symbolic Computation 95 (2019), 177–184. arXiv:1711.02628.

Companion documents in this repository: THE_WATERMARK_THEOREM (the surface order) ·
THE_DOUBLE_LADDER_THEOREM (the surface structure) · THE_BRAUER_COROLLARY ·
CHUCHIPACHI_FINDINGS_MASTER (the campaign record, graveyards included).